

Mathematics for Data Science

Reference Proof Sheet – Lecture 3

Determinants, Orthogonal Matrices, Eigenvalues and Diagonalization

M1 – AI / Data Science

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1 Permutations, transpositions and signatures

1.1 Counting permutations

Proposition
The number of permutations of $\{1, \dots, n\}$ is $\text{Card}(\mathcal{S}_n) = n!.$

Proof. To construct a permutation σ , we choose $\sigma(1)$ first. There are n possibilities. Once $\sigma(1)$ has been chosen, there are $n - 1$ possible values for $\sigma(2)$, because a permutation is injective. Continuing in this way, there are

$$n(n - 1)(n - 2) \cdots 2 \cdot 1 = n!$$

possible choices. Each such choice gives exactly one bijection from $\{1, \dots, n\}$ to itself, hence exactly one permutation. $\square$

1.2 Every permutation is a composition of transpositions

Proposition – Composition of transpositions
Every permutation $\sigma \in \mathcal{S}_n$ can be written as a composition of transpositions.

Proof. We prove the result by induction on n .

Base case. For $n = 1$, the only permutation is the identity. It can be viewed as a composition of zero transpositions. The result is therefore true.

Induction step. Assume the statement is true for permutations of $\{1, \dots, n - 1\}$, and let $\sigma \in \mathcal{S}_n$. Set

$$k = \sigma(n).$$

If $k = n$, then σ fixes n . Its restriction to $\{1, \dots, n - 1\}$ is a permutation of that set, so by the induction hypothesis this restriction is a composition of transpositions. The same composition represents σ on $\{1, \dots, n\}$.

Assume now that $k \neq n$. Let $\tau_{k,n}$ be the transposition exchanging k and n , and define

$$\rho = \tau_{k,n} \circ \sigma.$$

Then

$$\rho(n) = \tau_{k,n}(\sigma(n)) = \tau_{k,n}(k) = n.$$

Thus ρ fixes n , and its restriction to $\{1, \dots, n - 1\}$ is a permutation of that set. By the induction hypothesis, ρ is a composition of transpositions. Since $\tau_{k,n}^{-1} = \tau_{k,n}$,

$$\sigma = \tau_{k,n} \circ \rho.$$

Hence σ is also a composition of transpositions. $\square$

1.3 The signature

Definition – Signature of a permutation

For $\sigma \in \mathcal{S}_n$, define

$$\varepsilon(\sigma) = \prod_{1 \leq i < j \leq n} \frac{\sigma(j) - \sigma(i)}{j - i}.$$

Proposition – Signature and inversions

Let $N(\sigma)$ be the number of inversions of σ , that is, the number of pairs (i, j) such that

$$i < j \quad \text{and} \quad \sigma(i) > \sigma(j).$$

Then

$$\varepsilon(\sigma) = (-1)^{N(\sigma)}.$$

In particular,

$$\varepsilon(\sigma) \in \{-1, 1\}.$$

Proof. The denominator

$$\prod_{1 \leq i < j \leq n} (j - i)$$

is strictly positive.

Consider the numerator

$$\prod_{1 \leq i < j \leq n} (\sigma(j) - \sigma(i)).$$

Because σ is a permutation, the unordered pairs $\{\sigma(i), \sigma(j)\}$, with $i < j$, are exactly the unordered pairs of distinct elements of $\{1, \dots, n\}$. Therefore the absolute value of the numerator is equal to the denominator. Hence $|\varepsilon(\sigma)| = 1$.

The factor $\sigma(j) - \sigma(i)$ is negative exactly when $\sigma(i) > \sigma(j)$, that is, exactly when (i, j) is an inversion. Thus the numerator contains exactly $N(\sigma)$ negative factors. Consequently

$$\varepsilon(\sigma) = (-1)^{N(\sigma)}.$$

□

Lemma – Effect of a transposition on the signature

Let τ be a transposition and $\sigma \in \mathcal{S}_n$. Then

$$\varepsilon(\tau \circ \sigma) = -\varepsilon(\sigma).$$

In particular,

$$\varepsilon(\tau) = -1.$$

Proof. In the numerator of the definition of $\varepsilon(\sigma)$, the values $\sigma(1), \dots, \sigma(n)$ occur as the arguments of the following product (NB: this is called a *Vandermonde product*.)

$$\prod_{i < j} (x_j - x_i).$$

Composing on the left by a transposition τ exchanges two of the values among $\sigma(1), \dots, \sigma(n)$.

We show that exchanging two values in a Vandermonde product changes its sign. It is enough to consider an exchange of two adjacent values: all factors not involving these two values remain unchanged, while the factor formed by these two values changes sign, then the remaining affected factors are merely exchanged in pairs. Thus an adjacent exchange multiplies the product by -1 . Any transposition of two positions can be written as an odd number of adjacent exchanges, so it also multiplies the product by -1 . The denominator in the definition of ε is unchanged. Hence

$$\varepsilon(\tau \circ \sigma) = -\varepsilon(\sigma).$$

Taking $\sigma = \text{Id}$ and using $\varepsilon(\text{Id}) = 1$ gives

$$\varepsilon(\tau) = -1.$$

□

Proposition – Multiplicativity of the signature
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For all $\sigma, \pi \in \mathcal{S}_n$,
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$\varepsilon(\sigma \circ \pi) = \varepsilon(\sigma)\varepsilon(\pi).$
--

Proof. By the previous proposition, write

$$\sigma = \tau_1 \circ \cdots \circ \tau_T$$

for transpositions $\tau_1, \dots, \tau_T$. Repeatedly using the lemma above gives

$$\varepsilon(\sigma \circ \pi) = (-1)^T \varepsilon(\pi).$$

Applying the same argument to σ itself gives

$$\varepsilon(\sigma) = (-1)^T \varepsilon(\text{Id}) = (-1)^T.$$

Therefore

$$\varepsilon(\sigma \circ \pi) = \varepsilon(\sigma)\varepsilon(\pi).$$

□

Corollary

For every $\sigma \in \mathcal{S}_n$,

$\varepsilon(\sigma^{-1}) = \varepsilon(\sigma).$

Proof. Since $\sigma \circ \sigma^{-1} = \text{Id}$, multiplicativity gives

$$1 = \varepsilon(\text{Id}) = \varepsilon(\sigma)\varepsilon(\sigma^{-1}).$$

Both factors belong to $\{-1, 1\}$, hence

$$\varepsilon(\sigma^{-1}) = \varepsilon(\sigma).$$

□

2 The determinant and its structural properties

2.1 Leibniz formula

Definition – Determinant of a square matrix

Let $A = (a_{ij}) \in \mathcal{M}_n(\mathbb{K})$. Its determinant is

$$\det(A) = \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{j=1}^n a_{\sigma(j),j}.$$

Proposition

The determinant of the identity matrix is

$$\det(I_n) = 1.$$

Proof. In the Leibniz formula for I_n , the product

$$\prod_{j=1}^n (I_n)_{\sigma(j),j}$$

is nonzero only if $\sigma(j) = j$ for every j , that is, only for $\sigma = \text{Id}$. For the identity permutation, the product equals 1 and $\varepsilon(\text{Id}) = 1$. Thus

$$\det(I_n) = 1.$$

□

2.2 Multilinearity and alternation**Proposition – Multilinearity in the columns**

If all columns except column j are fixed, the determinant is linear with respect to column j . Equivalently, for column vectors $u, v \in \mathbb{K}^n$ and scalars $\alpha, \beta \in \mathbb{K}$,

$$\det(C_1, \dots, \alpha u + \beta v, \dots, C_n) = \alpha \det(C_1, \dots, u, \dots, C_n) + \beta \det(C_1, \dots, v, \dots, C_n).$$

Proof. In every term of the Leibniz formula, exactly one coefficient is selected from column j . If column j is $\alpha u + \beta v$, that coefficient is

$$\alpha u_{\sigma(j)} + \beta v_{\sigma(j)}.$$

Distributing this sum inside the Leibniz formula separates the determinant into two sums:

$$\begin{aligned} \det(C_1, \dots, \alpha u + \beta v, \dots, C_n) \\ = \alpha \det(C_1, \dots, u, \dots, C_n) + \beta \det(C_1, \dots, v, \dots, C_n). \end{aligned}$$

Thus the determinant is linear in column j . Since j was arbitrary, the determinant is multilinear in its columns. □

Proposition – Exchanging two columns

Exchanging two columns of a square matrix multiplies its determinant by -1 . Consequently, if two columns are equal, the determinant is zero.

Proof. Suppose that columns p and q are exchanged, and let τ be the transposition exchanging p and q . Call the resulting matrix B .

In the Leibniz formula for B , reindex the sum by the bijection $\sigma \mapsto \sigma \circ \tau$ of $\mathcal{S}_n$. The product of selected entries is then the same as the corresponding product for A , while

$$\varepsilon(\sigma \circ \tau) = \varepsilon(\sigma)\varepsilon(\tau) = -\varepsilon(\sigma).$$

Therefore

$$\det(B) = -\det(A).$$

If two columns of A are equal, exchanging them leaves A unchanged, but the previous result gives

$$\det(A) = -\det(A).$$

Hence $\det(A) = 0$. □

Lemma – Alternating multilinear forms change sign under an exchange
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Let $f : E^n \rightarrow \mathbb{K}$ be multilinear and alternating. If two arguments are exchanged, the value of f is multiplied by -1 .

Proof. It is enough to consider two positions while keeping all other arguments fixed. For $u, v \in E$, alternation gives

$$0 = f(\dots, u + v, \dots, u + v, \dots).$$

By multilinearity,

$$\begin{aligned} 0 &= f(\dots, u, \dots, u, \dots) + f(\dots, u, \dots, v, \dots) \\ &\quad + f(\dots, v, \dots, u, \dots) + f(\dots, v, \dots, v, \dots). \end{aligned}$$

The first and last terms are zero by alternation. Therefore

$$f(\dots, v, \dots, u, \dots) = -f(\dots, u, \dots, v, \dots).$$

□

2.3 Characterization by multilinearity, alternation and normalization

Proposition – Characterization of the determinant in a basis

Let E be an n -dimensional vector space over $\mathbb{K}$, and let

$$\mathcal{B} = (e_1, \dots, e_n)$$

be a basis of E . There exists a unique map

$$\det_{\mathcal{B}} : E^n \rightarrow \mathbb{K}$$

such that

- | |
|--|
| <ul style="list-style-type: none"> (i) $\det_{\mathcal{B}}$ is multilinear (ii) $\det_{\mathcal{B}}$ is alternating (iii) $\det_{\mathcal{B}}(e_1, \dots, e_n) = 1$. |
|--|

If $v_1, \dots, v_n \in E$ and X is the matrix whose j th column is $[v_j]_{\mathcal{B}}$, then

$$\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(X).$$

Proof.

Existence. For $v_1, \dots, v_n \in E$, let

$$X = ([v_1]_{\mathcal{B}} \cdots [v_n]_{\mathcal{B}}).$$

Define

$$\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(X).$$

Coordinates depend linearly on the vector, and the matrix determinant is multilinear and alternating in its columns. Therefore $\det_{\mathcal{B}}$ is multilinear and alternating. Moreover, the coordinate matrix of $(e_1, \dots, e_n)$ in the basis $\mathcal{B}$ is I_n , hence

$$\det_{\mathcal{B}}(e_1, \dots, e_n) = \det(I_n) = 1.$$

Thus such a map exists.

Uniqueness. Let $f : E^n \rightarrow \mathbb{K}$ be any multilinear alternating map satisfying $f(e_1, \dots, e_n) = 1$. Write

$$v_j = \sum_{i=1}^n a_{ij} e_i.$$

By multilinearity,

$$f(v_1, \dots, v_n) = \sum_{i_1=1}^n \cdots \sum_{i_n=1}^n \left(\prod_{j=1}^n a_{i_j j} \right) f(e_{i_1}, \dots, e_{i_n}).$$

If two indices among $i_1, \dots, i_n$ are equal, the corresponding term is zero because f is alternating. Thus only tuples $(i_1, \dots, i_n)$ in which every integer $1, \dots, n$ appears exactly once can contribute. Such tuples are precisely

$$(i_1, \dots, i_n) = (\sigma(1), \dots, \sigma(n)), \quad \sigma \in \mathcal{S}_n.$$

For such a permutation, exchanging basis vectors into their natural order gives

$$f(e_{\sigma(1)}, \dots, e_{\sigma(n)}) = \varepsilon(\sigma) f(e_1, \dots, e_n) = \varepsilon(\sigma).$$

Therefore

$$f(v_1, \dots, v_n) = \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{j=1}^n a_{\sigma(j), j} = \det(X).$$

Thus every map satisfying the three properties agrees with the determinant defined above, proving uniqueness. $\square$

2.4 Elementary column and row operations

Proposition – Elementary determinant rules for columns

Let A be a square matrix.

- (i) Exchanging two columns multiplies $\det(A)$ by -1 .
- (ii) Multiplying one column by λ multiplies $\det(A)$ by λ .
- (iii) Adding a multiple of one column to another column does not change $\det(A)$.

Proof. The first property was proved above.

For the second property, multilinearity in column j gives

$$\det(C_1, \dots, \lambda C_j, \dots, C_n) = \lambda \det(C_1, \dots, C_j, \dots, C_n).$$

For the third property, suppose $i \neq j$. By linearity in column j ,

$$\begin{aligned} \det(C_1, \dots, C_j + \lambda C_i, \dots, C_n) \\ = \det(C_1, \dots, C_j, \dots, C_n) + \lambda \det(C_1, \dots, C_i, \dots, C_n), \end{aligned}$$

where in the second determinant both columns i and j are equal to C_i . That determinant is zero by alternation. Hence adding λC_i to column j does not change the determinant. $\square$

2.5 The determinant of the transpose

Proposition

For every $A \in \mathcal{M}_n(\mathbb{K})$,

$$\det(A^\top) = \det(A).$$

Proof. Let $A = (a_{ij})$. By the Leibniz formula,

$$\det(A^\top) = \sum_{\sigma \in \mathcal{S}_n} \varepsilon(\sigma) \prod_{j=1}^n a_{j, \sigma(j)}.$$

Set $\tau = \sigma^{-1}$. Since inversion is a bijection of $\mathcal{S}_n$ and $\varepsilon(\sigma^{-1}) = \varepsilon(\sigma)$,

$$\det(A^\top) = \sum_{\tau \in \mathcal{S}_n} \varepsilon(\tau) \prod_{j=1}^n a_{j, \tau^{-1}(j)}.$$

Reindex the product by $k = \tau^{-1}(j)$, so that $j = \tau(k)$. Then

$$\prod_{j=1}^n a_{j, \tau^{-1}(j)} = \prod_{k=1}^n a_{\tau(k), k}.$$

Therefore

$$\det(A^\top) = \sum_{\tau \in \mathcal{S}_n} \varepsilon(\tau) \prod_{k=1}^n a_{\tau(k), k} = \det(A).$$

$\square$

Corollary – Elementary determinant rules for rows
Exchanging two rows multiplies the determinant by -1 , multiplying one row by λ multiplies the determinant by λ , adding a multiple of one row to another row leaves the determinant unchanged.

Proof. Row operations on A are the corresponding column operations on $A^\top$. Apply the column-operation rules to $A^\top$ and use

$$\det(A^\top) = \det(A).$$

□

2.6 Determinant of a product

Proposition – Multiplicativity of the determinant
For $A, B \in \mathcal{M}_n(\mathbb{K})$,
$\det(AB) = \det(A) \det(B).$

Proof. Fix A and define a function of n column vectors by

$$f(C_1, \dots, C_n) = \det(AC_1, \dots, AC_n).$$

Because multiplication by A is linear and the determinant is multilinear, f is multilinear. If two arguments C_i and C_j are equal, then $AC_i = AC_j$, so $f = 0$. Thus f is alternating.

By the characterization of alternating multilinear forms relative to the canonical basis $(e_1, \dots, e_n)$, there exists a scalar c such that

$$f(C_1, \dots, C_n) = c \det(C_1, \dots, C_n).$$

To determine c , evaluate at the canonical basis:

$$c = f(e_1, \dots, e_n) = \det(Ae_1, \dots, Ae_n) = \det(A),$$

because $Ae_1, \dots, Ae_n$ are the columns of A . Hence

$$f(C_1, \dots, C_n) = \det(A) \det(C_1, \dots, C_n).$$

Taking $C_1, \dots, C_n$ to be the columns of B , the columns of AB are $AC_1, \dots, AC_n$. Therefore

$$\det(AB) = \det(A) \det(B).$$

□

Corollary – Scaling a whole matrix
For $\alpha \in \mathbb{K}$ and $A \in \mathcal{M}_n(\mathbb{K})$,
$\det(\alpha A) = \alpha^n \det(A).$

Proof. Multiplying the whole matrix by α multiplies each of its n columns by α . By multilinearity,

$$\det(\alpha C_1, \dots, \alpha C_n) = \alpha^n \det(C_1, \dots, C_n).$$

□

3 Laplace expansion

3.1 Expansion along a column

Proposition – Laplace expansion along column j

Let $A = (a_{ij}) \in \mathcal{M}_n(\mathbb{K})$. Let M_{ij} be the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j from A . Then

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(M_{ij}).$$

Equivalently, if

$$C_{ij} = (-1)^{i+j} \det(M_{ij})$$

is the cofactor of a_{ij} , then

$$\det(A) = \sum_{i=1}^n a_{ij} C_{ij}.$$

Proof. Fix the column index j . Starting from the Leibniz formula, group the permutations according to the value $i = \sigma(j)$:

$$\det(A) = \sum_{i=1}^n a_{ij} \sum_{\substack{\sigma \in \mathcal{S}_n \\ \sigma(j)=i}} \varepsilon(\sigma) \prod_{k \neq j} a_{\sigma(k),k}.$$

We identify the inner sum.

Delete position j from the domain and value i from the range. After relabeling both remaining ordered sets increasingly by $\{1, \dots, n-1\}$, each permutation σ satisfying $\sigma(j) = i$ corresponds to exactly one permutation $\pi \in \mathcal{S}_{n-1}$. The product

$$\prod_{k \neq j} a_{\sigma(k),k}$$

is then exactly the product associated with π in the Leibniz formula for M_{ij} .

It remains to compare the signs. Let $N(\sigma)$ and $N(\pi)$ denote the numbers of inversions. The inversions of σ not involving position j are exactly the inversions of π . Let

$$a = \#\{k < j : \sigma(k) > i\}, \quad b = \#\{k > j : \sigma(k) < i\}.$$

These are precisely the inversions involving position j . If

$$c = \#\{k < j : \sigma(k) < i\},$$

then

$$a + c = j - 1, \quad b + c = i - 1.$$

Hence

$$a + b = i + j - 2 - 2c,$$

so $a + b$ has the same parity as $i + j$. Therefore

$$\varepsilon(\sigma) = (-1)^{N(\sigma)} = (-1)^{i+j} (-1)^{N(\pi)} = (-1)^{i+j} \varepsilon(\pi).$$

Thus

$$\sum_{\substack{\sigma \in \mathcal{S}_n \\ \sigma(j)=i}} \varepsilon(\sigma) \prod_{k \neq j} a_{\sigma(k),k} = (-1)^{i+j} \det(M_{ij}).$$

Substituting this into the grouped Leibniz formula gives

$$\det(A) = \sum_{i=1}^n (-1)^{i+j} a_{ij} \det(M_{ij}).$$

□

3.2 Expansion along a row

Proposition – Laplace expansion along row i

For a fixed row i ,

$$\det(A) = \sum_{j=1}^n (-1)^{i+j} a_{ij} \det(M_{ij}) = \sum_{j=1}^n a_{ij} C_{ij}.$$

Proof. Apply the column expansion to $A^\top$ along column i . The minor of $(A^\top)_{ji} = a_{ij}$ is $M_{ij}^\top$, whose determinant equals $\det(M_{ij})$. Using $\det(A^\top) = \det(A)$ gives

$$\det(A) = \sum_{j=1}^n (-1)^{j+i} a_{ij} \det(M_{ij}),$$

which is the required row expansion. □

4 Determinant, bases, change of basis and invertibility

4.1 Change of basis for the determinant form

Proposition – Basis change and determinant

Let $\mathcal{B} = (e_1, \dots, e_n)$ and $\mathcal{B}' = (e'_1, \dots, e'_n)$ be two bases of an n -dimensional vector space E . Let P be the change-of-basis matrix whose j th column is $[e'_j]_{\mathcal{B}}$, so that

$$[x]_{\mathcal{B}} = P[x]_{\mathcal{B}'}$$

Then for every $v_1, \dots, v_n \in E$,

$$\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(P) \det_{\mathcal{B}'}(v_1, \dots, v_n).$$

Equivalently,

$$\det_{\mathcal{B}'}(v_1, \dots, v_n) = \frac{1}{\det(P)} \det_{\mathcal{B}}(v_1, \dots, v_n).$$

Proof. Let

$$X = ([v_1]_{\mathcal{B}} \cdots [v_n]_{\mathcal{B}}), \quad X' = ([v_1]_{\mathcal{B}'} \cdots [v_n]_{\mathcal{B}'}).$$

For each j ,

$$[v_j]_{\mathcal{B}} = P[v_j]_{\mathcal{B}'}$$

Therefore, column by column,

$$X = PX'.$$

Taking determinants and using multiplicativity,

$$\det(X) = \det(P) \det(X').$$

By definition of the determinant forms,

$$\det(X) = \det_{\mathcal{B}}(v_1, \dots, v_n), \quad \det(X') = \det_{\mathcal{B}'}(v_1, \dots, v_n).$$

This proves the first formula.

The matrix P is invertible because it is a change-of-basis matrix. Hence

$$1 = \det(I_n) = \det(PP^{-1}) = \det(P) \det(P^{-1}),$$

so $\det(P) \neq 0$. Dividing by $\det(P)$ gives the equivalent formula. $\square$

4.2 A family is a basis if and only if its determinant is nonzero

Proposition

Let E be n -dimensional, let $\mathcal{B}$ be a basis of E , and let $v_1, \dots, v_n \in E$. Then

$$(v_1, \dots, v_n) \text{ is a basis of } E \iff \det_{\mathcal{B}}(v_1, \dots, v_n) \neq 0.$$

Proof.

If the family is linearly dependent, its determinant is zero. Assume $(v_1, \dots, v_n)$ is linearly dependent. Then one vector, say v_j , is a linear combination of the others:

$$v_j = \sum_{k \neq j} \alpha_k v_k.$$

By multilinearity,

$$\det_{\mathcal{B}}(v_1, \dots, v_j, \dots, v_n) = \sum_{k \neq j} \alpha_k \det_{\mathcal{B}}(v_1, \dots, v_k, \dots, v_n).$$

In each determinant on the right, the vector v_k appears twice, so every term is zero by alternation. Hence

$$\det_{\mathcal{B}}(v_1, \dots, v_n) = 0.$$

By contraposition,

$$\det_{\mathcal{B}}(v_1, \dots, v_n) \neq 0 \implies (v_1, \dots, v_n) \text{ is linearly independent.}$$

Since E has dimension n , n linearly independent vectors form a basis. This proves one implication.

If the family is a basis, its determinant is nonzero. Assume $\mathcal{B}' = (v_1, \dots, v_n)$ is a basis. The change-of-basis formula gives

$$\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(P) \det_{\mathcal{B}'}(v_1, \dots, v_n).$$

Since $(v_1, \dots, v_n)$ is precisely the basis $\mathcal{B}'$,

$$\det_{\mathcal{B}'}(v_1, \dots, v_n) = 1.$$

Also P is invertible, so as shown above $\det(P) \neq 0$. Therefore

$$\det_{\mathcal{B}}(v_1, \dots, v_n) = \det(P) \neq 0.$$

$\square$

4.3 Invertibility criterion

Theorem – Determinant and invertibility	
For $A \in \mathcal{M}_n(\mathbb{K})$,	$A \text{ is invertible} \iff \det(A) \neq 0.$
Equivalently,	$A \text{ is singular} \iff \det(A) = 0.$

Proof. Let $C_1, \dots, C_n$ be the columns of A . By Lecture 2, a square matrix is invertible if and only if its columns form a basis of $\mathbb{K}^n$. By the proposition above, the columns form a basis if and only if

$$\det(C_1, \dots, C_n) = \det(A) \neq 0.$$

Thus

$$A \text{ invertible} \iff \det(A) \neq 0.$$

The statement about singular matrices is the negation of this equivalence. $\square$

Corollary – Determinant of the inverse	
If A is invertible, then	$\det(A^{-1}) = \frac{1}{\det(A)}.$

Proof. Since $AA^{-1} = I_n$,

$$1 = \det(I_n) = \det(AA^{-1}) = \det(A) \det(A^{-1}).$$

Since A is invertible, $\det(A) \neq 0$, so

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

$\square$

5 Determinant of an endomorphism

5.1 Similarity invariance

Proposition – Determinant is invariant under similarity	
Let $A, B \in \mathcal{M}_n(\mathbb{K})$ and suppose	$B = P^{-1}AP$
for some invertible matrix P . Then	$\det(B) = \det(A).$

Proof. Using multiplicativity of the determinant,

$$\begin{aligned}\det(B) &= \det(P^{-1}AP) \\ &= \det(P^{-1}) \det(A) \det(P) \\ &= \frac{1}{\det(P)} \det(A) \det(P) \\ &= \det(A).\end{aligned}$$

□

Definition – Determinant of an endomorphism

Let $f : E \rightarrow E$ be an endomorphism of a finite-dimensional vector space. If $A = [f]_{\mathcal{B}}$ is the matrix of f in any basis $\mathcal{B}$, define

$$\det(f) = \det(A).$$

This definition does not depend on the chosen basis.

Proof. Let $\mathcal{B}$ and $\mathcal{B}'$ be two bases, and let

$$A = [f]_{\mathcal{B}}, \quad B = [f]_{\mathcal{B}}.$$

By the change-of-basis theorem from Lecture 2, there exists an invertible matrix P such that

$$B = P^{-1}AP.$$

Similarity invariance gives

$$\det(B) = \det(A).$$

Hence the determinant of the matrix representing f is independent of the basis. □

5.2 Basic properties for endomorphisms

Proposition

Let $f, g : E \rightarrow E$ be endomorphisms of a finite-dimensional vector space E . Then:

- (i) f is bijective if and only if $\det(f) \neq 0$;
- (ii) $\det(f \circ g) = \det(f) \det(g)$;
- (iii) for every basis $\mathcal{B} = (v_1, \dots, v_n)$ of E ,

$$\det_{\mathcal{B}}(f(v_1), \dots, f(v_n)) = \det(f) \det_{\mathcal{B}}(v_1, \dots, v_n);$$

- (iv) if f is bijective, then

$$\det(f^{-1}) = \frac{1}{\det(f)}.$$

Proof. Fix a basis $\mathcal{B}$ and write

$$A = [f]_{\mathcal{B}}, \quad B = [g]_{\mathcal{B}}.$$

Bijectivity. The map f is bijective if and only if its matrix A is invertible. By the determinant criterion,

$$A \text{ invertible} \iff \det(A) \neq 0.$$

Since $\det(f) = \det(A)$, this proves (i).

Composition. The matrix of $f \circ g$ in $\mathcal{B}$ is AB . Therefore

$$\det(f \circ g) = \det(AB) = \det(A) \det(B) = \det(f) \det(g).$$

Action on a basis. Since $\mathcal{B} = (v_1, \dots, v_n)$, its coordinate matrix in itself is I_n . The coordinate matrix of

$$(f(v_1), \dots, f(v_n))$$

in $\mathcal{B}$ is exactly A , because the columns of A are the coordinates of the images of the basis vectors. Hence

$$\det_{\mathcal{B}}(f(v_1), \dots, f(v_n)) = \det(A) = \det(f).$$

Also

$$\det_{\mathcal{B}}(v_1, \dots, v_n) = 1,$$

so the claimed formula follows.

Inverse. If f is bijective, then

$$f \circ f^{-1} = \text{Id}_E.$$

Thus

$$1 = \det(\text{Id}_E) = \det(f) \det(f^{-1}),$$

and therefore

$$\det(f^{-1}) = \frac{1}{\det(f)}.$$

□

Remark – A slightly more general form of (iii)
For arbitrary vectors $x_1, \dots, x_n \in E$, not necessarily a basis, $\det_{\mathcal{B}}(f(x_1), \dots, f(x_n)) = \det(f) \det_{\mathcal{B}}(x_1, \dots, x_n).$

Indeed, let X be the matrix whose j th column is $[x_j]_{\mathcal{B}}$. The j th column of AX is $[f(x_j)]_{\mathcal{B}}$. Hence the coordinate matrix of the transformed family is AX , and

$$\det(AX) = \det(A) \det(X).$$

6 Determinants of particular matrices

6.1 Diagonal matrices

Proposition – Determinant of a diagonal matrix
For $D = \text{diag}(\lambda_1, \dots, \lambda_n),$

we have

$$\det(D) = \prod_{i=1}^n \lambda_i.$$

Proof. In the Leibniz formula, a term

$$\prod_{j=1}^n d_{\sigma(j),j}$$

can be nonzero only if $\sigma(j) = j$ for every j , because all off-diagonal entries of D are zero. Thus only the identity permutation contributes, and

$$\det(D) = \varepsilon(\text{Id}) \prod_{j=1}^n d_{jj} = \prod_{j=1}^n \lambda_j.$$

□

Corollary

A diagonal matrix

$$D = \text{diag}(\lambda_1, \dots, \lambda_n)$$

is invertible if and only if every diagonal entry λ_i is nonzero.

Proof. By the determinant criterion,

$$D \text{ invertible} \iff \det(D) \neq 0.$$

But

$$\det(D) = \lambda_1 \cdots \lambda_n,$$

and a finite product of scalars is nonzero if and only if every factor is nonzero. □

6.2 Triangular matrices

Proposition – Determinant of a triangular matrix

If $T = (t_{ij}) \in \mathcal{M}_n(\mathbb{K})$ is upper or lower triangular, then

$$\det(T) = \prod_{i=1}^n t_{ii}.$$

Proof. We first consider an upper triangular matrix and prove the result by induction on n .

Base case. For $n = 1$, the statement is immediate.

Induction step. Assume the result holds for upper triangular matrices of size $(n-1) \times (n-1)$. If T is upper triangular, its first column has the form

$$\begin{pmatrix} t_{11} \\ 0 \\ \vdots \\ 0 \end{pmatrix}.$$

Expanding the determinant along the first column gives

$$\det(T) = t_{11} \det(M_{11}).$$

The minor M_{11} is again upper triangular, with diagonal entries $t_{22}, \dots, t_{nn}$. By the induction hypothesis,

$$\det(M_{11}) = t_{22} \cdots t_{nn}.$$

Therefore

$$\det(T) = t_{11} t_{22} \cdots t_{nn}.$$

If T is lower triangular, then $T^\top$ is upper triangular and has the same diagonal entries. Using $\det(T) = \det(T^\top)$ gives the same formula. $\square$

6.3 Orthogonal matrices

Proposition – Determinant of an orthogonal matrix
--

Let $U \in \mathbb{R}^{n \times n}$ be orthogonal, that is,

$$U^\top U = I_n.$$

Then

$$\det(U) \in \{-1, 1\}.$$

Proof. Taking determinants in

$$U^\top U = I_n$$

gives

$$\det(U^\top) \det(U) = \det(I_n) = 1.$$

Since $\det(U^\top) = \det(U)$,

$$(\det(U))^2 = 1.$$

Over $\mathbb{R}$, this implies

$$\det(U) = 1 \quad \text{or} \quad \det(U) = -1.$$

$\square$

Proposition – Columns and rows of an orthogonal matrix

If $U \in \mathbb{R}^{n \times n}$ is orthogonal, then its columns form an orthonormal basis of $\mathbb{R}^n$. Its rows also form an orthonormal basis of $\mathbb{R}^n$.

Proof. Write the columns of U as

$$U = (u_1 \cdots u_n).$$

The (i, j) entry of $U^\top U$ is

$$(U^\top U)_{ij} = u_i^\top u_j = u_i \cdot u_j.$$

Since $U^\top U = I_n$,

$$u_i \cdot u_j = \delta_{ij}.$$

Thus

$$\|u_i\|_2 = 1 \quad \text{for every } i, \quad u_i \cdot u_j = 0 \quad \text{for } i \neq j.$$

Hence the columns are orthonormal. There are n of them in $\mathbb{R}^n$, so they form an orthonormal basis.

Moreover, $U^\top U = I_n$ implies $U^{-1} = U^\top$, hence also

$$UU^\top = I_n.$$

Applying the same argument to the rows of U proves that they are orthonormal as well. $\square$

Proposition – Orthogonal matrices preserve inner products and norms
--

If $Q \in \mathbb{R}^{n \times n}$ is orthogonal, then for all $x, y \in \mathbb{R}^n$,
--

$(Qx) \cdot (Qy) = x \cdot y.$

Consequently,

$\ Qx\ _2 = \ x\ _2.$

Proof. Since $Q^\top Q = I_n$,

$$\begin{aligned}
 (Qx) \cdot (Qy) &= (Qx)^\top (Qy) \\
 &= x^\top Q^\top Qy \\
 &= x^\top y \\
 &= x \cdot y.
 \end{aligned}$$

Taking $y = x$ gives

$$\|Qx\|_2^2 = (Qx) \cdot (Qx) = x \cdot x = \|x\|_2^2.$$

Both norms are nonnegative, so

$$\|Qx\|_2 = \|x\|_2.$$

$\square$

Example – Rotation matrix

For

$R(\theta) = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix},$
--

we have

$R(\theta)^\top R(\theta) = I_2, \quad \det(R(\theta)) = 1, \quad R(\theta)^{-1} = R(-\theta).$

Indeed,

$$R(\theta)^\top R(\theta) = \begin{pmatrix} \cos^2 \theta + \sin^2 \theta & 0 \\ 0 & \cos^2 \theta + \sin^2 \theta \end{pmatrix} = I_2,$$

and

$$\det(R(\theta)) = \cos^2 \theta + \sin^2 \theta = 1.$$

Finally,

$$R(\theta)^\top = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} = R(-\theta),$$

so orthogonality gives $R(\theta)^{-1} = R(-\theta)$.

7 Eigenvalues, eigenspaces and diagonalization

7.1 Eigenspaces are subspaces

Proposition – Eigenspace

Let $A \in \mathbb{R}^{n \times n}$ and let $\lambda \in \mathbb{R}$ be an eigenvalue of A . The associated eigenspace is

$$E_\lambda = \{v \in \mathbb{R}^n : Av = \lambda v\} = \ker(A - \lambda I_n).$$

It is a vector subspace of $\mathbb{R}^n$.

Proof. For $v \in \mathbb{R}^n$,

$$\begin{aligned} Av = \lambda v &\iff Av - \lambda v = 0 \\ &\iff (A - \lambda I_n)v = 0. \end{aligned}$$

Thus

$$E_\lambda = \ker(A - \lambda I_n).$$

The map

$$x \mapsto (A - \lambda I_n)x$$

is linear. By Lecture 1, the kernel of a linear map is a vector subspace. Therefore E_λ is a subspace of $\mathbb{R}^n$. $\square$

7.2 Diagonalization from an eigenbasis

Proposition – Diagonalization from linearly independent eigenvectors

Let $A \in \mathbb{R}^{n \times n}$. Suppose $v_1, \dots, v_n$ are linearly independent eigenvectors of A , with

$$Av_i = \lambda_i v_i, \quad i = 1, \dots, n.$$

Then $(v_1, \dots, v_n)$ is a basis of $\mathbb{R}^n$ and A is diagonalizable. More precisely, if

$$V = (v_1 \ \cdots \ v_n), \quad D = \text{diag}(\lambda_1, \dots, \lambda_n),$$

then

$$AV = VD$$

and

$$A = VDV^{-1}.$$

Proof. Because $v_1, \dots, v_n$ are n linearly independent vectors in the n -dimensional space $\mathbb{R}^n$, they form a basis. Therefore the matrix

$$V = (v_1 \ \cdots \ v_n)$$

is invertible.

Now multiply A by V column by column:

$$AV = (Av_1 \ \cdots \ Av_n) = (\lambda_1 v_1 \ \cdots \ \lambda_n v_n).$$

On the other hand, right multiplication of V by the diagonal matrix D multiplies the i th column of V by λ_i . Hence

$$VD = (\lambda_1 v_1 \ \cdots \ \lambda_n v_n).$$

Thus

$$AV = VD.$$

Since V is invertible, multiplying on the right by V^{-1} gives

$$A = VDV^{-1}.$$

Therefore A is diagonalizable. □

Remark – Meaning of the diagonalization
The identity $A = VDV^{-1}$ expresses the same linear transformation in two coordinate systems: D is the matrix of the transformation in the eigenbasis $(v_1, \dots, v_n)$, while A is its matrix in the canonical basis.

7.3 Powers of a diagonalizable matrix

Proposition – Eigenvectors and powers
Assume $Av_i = \lambda_i v_i, \quad i = 1, \dots, n,$ where $v_1, \dots, v_n$ are linearly independent. Then for every integer $k \geq 1$, $A^k v_i = \lambda_i^k v_i.$ Hence the same vectors v_i are eigenvectors of A^k, with associated eigenvalues λ_i^k. Moreover, $A^k = VD^k V^{-1} = V \operatorname{diag}(\lambda_1^k, \dots, \lambda_n^k) V^{-1},$ so every A^k is diagonalizable.

Proof.

Action on each eigenvector. We prove by induction on k that

$$A^k v_i = \lambda_i^k v_i.$$

For $k = 1$, this is exactly the eigenvector equation. Assume it holds for some $k \geq 1$. Then

$$\begin{aligned} A^{k+1} v_i &= A(A^k v_i) \\ &= A(\lambda_i^k v_i) \\ &= \lambda_i^k A v_i \\ &= \lambda_i^k \lambda_i v_i \\ &= \lambda_i^{k+1} v_i. \end{aligned}$$

Thus the formula holds for every $k \geq 1$.

Matrix formula. Since

$$A = VDV^{-1},$$

we have

$$\begin{aligned} A^2 &= (VDV^{-1})(VDV^{-1}) \\ &= VD(V^{-1}V)DV^{-1} \\ &= VD^2V^{-1}. \end{aligned}$$

Repeating the cancellation $V^{-1}V = I_n$ gives, by induction,

$$A^k = VD^kV^{-1}.$$

Because D is diagonal,

$$D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k).$$

The same invertible matrix V therefore diagonalizes every A^k . □

Remark – The case $k = 0$
For $k = 0$, $A^0 = I_n = VI_nV^{-1}.$ Every nonzero vector is an eigenvector of I_n with eigenvalue 1; in particular, the vectors $v_1, \dots, v_n$ remain an eigenbasis.